Interval search experiments

Built 2026-10-16 20:33 PT. Experiment design by Roland Gao; runs and analysis by Claude. Code: research/cooldown_search_probe.py (runs), research/cooldown_study.py (analysis), research/make_cooldown_study.py (this PDF). Logs: logs/research_cooldown_search_probe_*, logs/research_hparam_importance_*. Batch size 2000 throughout.

Experiment 1. Cooldown study: which 40-step lookahead ranks interval candidates like the final result?

Question

An interval search picks each interval's LR by training the interval with a candidate LR, then a short lookahead (the cooldown), and scoring the result. Which cooldown ranks the candidates the way the final accuracy does? Compared, all with a 40-step lookahead (one interval at batch size 2000): no lookahead; linear decay to 0 (conv only, or all four groups); one searched constant cooldown, either for conv only with the other groups on their ground-truth schedules, or shared by all four groups; and four searched constant cooldowns (one per group).

Setup

Item	Value
Model / data	CIFAR-10 airbench, batch size 2000, 200 steps, baseline_bs2000 (conv Muon2 0.22 -> 0; head, norm_bias, whiten_bias SGD on their baseline schedules)
Intervals	4 intervals of 40 steps: steps 0-40, 40-80, 80-120, 120-160 (the last interval has no room for a 40-step lookahead)
Candidates	7 constant conv LRs for the interval: the 0.6^k grid point nearest the baseline's mean conv LR in that interval, and 3 grid steps either side; other groups on the baseline
Ground truth	Final test TTA when training continues on the baseline schedule after the interval
No lookahead	Test TTA right after the interval
Linear to 0	Conv only: conv decays linearly from the candidate to 0 over 40 steps, the other groups continue their schedules. All groups: every group decays linearly to 0 from its current value
1 searched cooldown: conv only	Conv constant at m x candidate for 40 steps; the other groups follow their baseline (ground-truth) schedules; m chosen from {1, 0.6, 0.36, 0.22, 0.13, 0.078, 0.047, 0} by the best score on that seed
1 searched cooldown: shared	Constant LRs for 40 steps: conv at m x candidate, each other group at m x its mean LR over the interval; one m for all groups, chosen as above
4 searched cooldowns	As above but one m per group; coordinate search from the best shared m, one group at a time over the same 8 values, until no group changes (at most 3 passes)
Seed	1 training seed (0); every method uses the same prefix and data order
Score	Test-set TTA accuracy (%)

Results

Picked LR = the candidate with the highest score. Regret = ground truth at its best candidate minus ground truth at the picked one (pp). One seed: single-run noise is about 0.13 pp, so differences of that size between candidates are not resolved.

Picked LR (regret, pp)

Method	Interval 1	Interval 2	Interval 3	Interval 4
Ground truth (final TTA)	0.078 (0.00)	0.13 (0.00)	0.078 (0.00)	0.078 (0.00)
No lookahead	0.078 (0.00)	0.028 (0.34)	0.028 (0.01)	0.028 (0.29)
Linear to 0: conv only	0.13 (0.05)	0.078 (0.26)	0.078 (0.00)	0.028 (0.29)
Linear to 0: all four groups	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.028 (0.29)
1 searched cooldown: conv only, others on ground truth	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.047 (0.21)
1 searched cooldown: shared by all four groups	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.047 (0.21)
4 searched cooldowns (one per group)	0.13 (0.05)	0.078 (0.26)	0.047 (0.21)	0.047 (0.21)

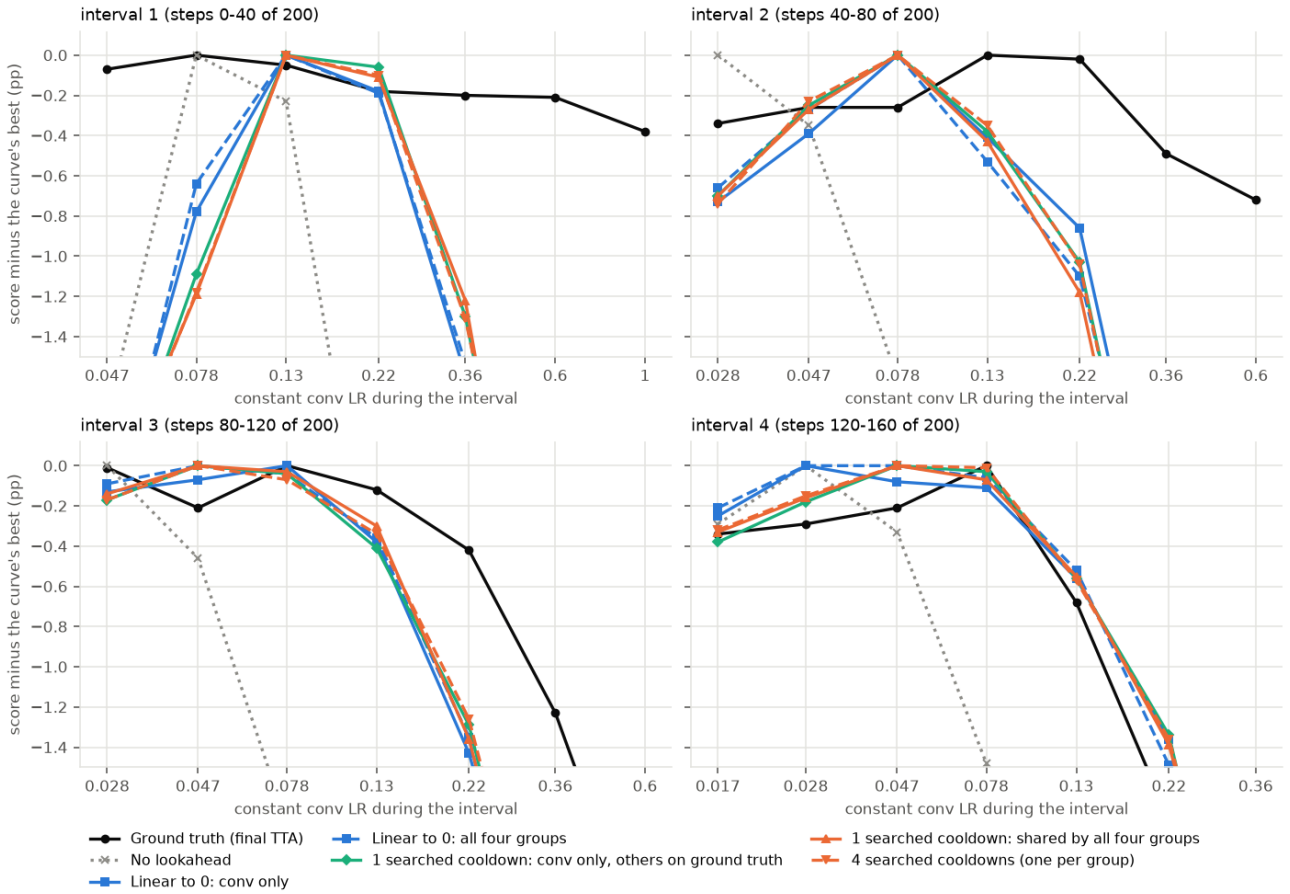


Figure 1. Each method's score per candidate, relative to its own best (the point at 0 is its pick).

Interval 1: score per candidate conv LR (bold: each row's best)

Method	0.047	0.078	0.13	0.22	0.36	0.6	1
Ground truth (final TTA)	93.99	**94.06**	94.01	93.88	93.86	93.85	93.68
No lookahead	81.57	**83.35**	83.12	80.45	68.80	56.07	35.18
Linear to 0: conv only	89.04	90.59	**91.37**	91.19	89.78	87.38	84.39
Linear to 0: all four groups	88.85	90.63	**91.27**	91.08	89.74	87.41	84.10
1 searched cooldown: conv only, others on ground truth	88.58	89.83	**90.92**	90.86	89.62	87.45	85.95
1 searched cooldown: shared by all four groups	88.65	89.70	**90.89**	90.78	89.67	87.16	85.13
4 searched cooldowns (one per group)	88.79	89.89	**91.07**	90.97	89.76	87.65	86.13

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.047	0.6 x1	0.6 x1	(1, 1, 0.6, 0.36) x1
0.078	0.36 x1	0.36 x1	(0.36, 1, 1, 0.22) x1
0.13	0.22 x1	0.22 x1	(0.22, 1, 0.22, 0.22) x1
0.22	0.22 x1	0.22 x1	(0.22, 0.36, 0.22, 1) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.047, 0.13, 0.13) x1
0.6	0.078 x1	0.078 x1	(0.078, 1, 0.36, 0.078) x1
1	0.078 x1	0.078 x1	(0.078, 0.6, 0.078, 0.6) x1

Interval 2: score per candidate conv LR (bold: each row's best)

Method	0.028	0.047	0.078	0.13	0.22	0.36	0.6
Ground truth (final TTA)	93.76	93.84	93.84	**94.10**	94.08	93.61	93.38
No lookahead	**91.12**	90.77	89.51	87.43	82.45	71.27	57.08
Linear to 0: conv only	92.25	92.59	**92.98**	92.57	92.12	90.18	87.94
Linear to 0: all four groups	92.27	92.66	**92.93**	92.40	91.83	90.11	87.78
1 searched cooldown: conv only, others on ground truth	92.16	92.61	**92.86**	92.48	91.83	89.82	87.93
1 searched cooldown: shared by all four groups	92.18	92.61	**92.88**	92.45	91.70	89.73	87.84
4 searched cooldowns (one per group)	92.20	92.71	**92.94**	92.59	91.90	89.87	87.98

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.028	0.6 x1	0.6 x1	(0.6, 0.6, 1, 0.6) x1
0.047	0.36 x1	0.36 x1	(0.36, 1, 0.047, 0.36) x1
0.078	0.22 x1	0.22 x1	(0.22, 0.13, 0.22, 0.22) x1
0.13	0.22 x1	0.22 x1	(0.22, 1, 0.36, 0) x1
0.22	0.13 x1	0.13 x1	(0.13, 1, 1, 0.078) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.6, 0, 0.047) x1
0.6	0.078 x1	0.078 x1	(0.078, 0.047, 0.13, 0.078) x1

Interval 3: score per candidate conv LR (bold: each row's best)

Method	0.028	0.047	0.078	0.13	0.22	0.36	0.6
Ground truth (final TTA)	93.99	93.79	**94.00**	93.88	93.58	92.77	91.54
No lookahead	**92.59**	92.13	90.81	87.68	82.32	71.78	56.20
Linear to 0: conv only	93.26	93.32	**93.39**	93.01	91.96	90.45	87.63
Linear to 0: all four groups	93.28	**93.37**	93.33	93.01	92.02	90.23	87.59
1 searched cooldown: conv only, others on ground truth	93.16	**93.33**	93.29	92.92	92.04	90.31	88.61
1 searched cooldown: shared by all four groups	93.13	**93.27**	93.24	92.97	91.91	90.29	88.64
4 searched cooldowns (one per group)	93.18	**93.35**	93.28	93.01	92.09	90.44	88.76

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.028	0.36 x1	0.22 x1	(0.22, 0.13, 1, 0.22) x1
0.047	0.22 x1	0.36 x1	(0.22, 1, 1, 0.36) x1
0.078	0.22 x1	0.22 x1	(0.22, 1, 0.36, 0.22) x1
0.13	0.22 x1	0.22 x1	(0.22, 1, 0.22, 0.22) x1
0.22	0.22 x1	0.13 x1	(0.22, 0.36, 1, 0.13) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.078, 0.6, 0.13) x1
0.6	0.078 x1	0.13 x1	(0.078, 0.36, 0.13, 0.13) x1

Interval 4: score per candidate conv LR (bold: each row's best)

Method	0.017	0.028	0.047	0.078	0.13	0.22	0.36
Ground truth (final TTA)	93.65	93.70	93.78	**93.99**	93.31	92.23	89.58
No lookahead	92.89	**93.18**	92.85	91.70	88.84	83.30	72.23
Linear to 0: conv only	93.55	**93.80**	93.72	93.69	93.24	92.44	90.63
Linear to 0: all four groups	93.55	**93.76**	**93.76**	93.70	93.24	92.27	90.57
1 searched cooldown: conv only, others on ground truth	93.51	93.71	**93.89**	93.86	93.33	92.55	90.70
1 searched cooldown: shared by all four groups	93.56	93.73	**93.89**	93.82	93.34	92.50	90.89
4 searched cooldowns (one per group)	93.62	93.79	**93.94**	93.93	93.37	92.58	90.93

Candidate	Conv only: chosen m (seeds)	Shared: chosen m (seeds)	4 searched: chosen (conv, head, norm_bias, whiten_bias) (seeds)
0.017	0.6 x1	0.6 x1	(0.6, 1, 1, 0.6) x1
0.028	0.6 x1	0.6 x1	(0.6, 1, 0.36, 0.6) x1
0.047	0.36 x1	0.36 x1	(0.36, 0.22, 1, 0.36) x1
0.078	0.22 x1	0.22 x1	(0.22, 0.047, 0.22, 0.22) x1
0.13	0.13 x1	0.13 x1	(0.13, 0.22, 0.13, 0.13) x1
0.22	0.13 x1	0.13 x1	(0.13, 1, 0.078, 0.13) x1
0.36	0.13 x1	0.13 x1	(0.13, 0.078, 1, 0.13) x1

Observations

One seed: single-run noise is about 0.13 pp, so ground-truth differences of that size are not resolved. Interval 1's ground truth is flat (93.85-94.06 from 0.047 to 0.6) and interval 3's is non-monotone (0.028: 93.99, 0.047: 93.79, 0.078: 94.00), so regrets of about 0.2 pp or less at those intervals are within noise.

All five 40-step lookahead variants pick the same LR at intervals 1 and 2, and the three searched-constant variants (conv only, shared, four per group) pick the same LR at every interval. How the other three groups cool down (continuing their ground-truth schedules, one shared multiplier, or their own multipliers) never changed the pick. Four searched multipliers score slightly higher in absolute terms than one but rank the candidates the same way.

The 40-step lookahead picks a smaller LR than the ground truth where the ground truth is clearly peaked: interval 2 (all variants pick 0.078 where 0.13 is best, 0.26 pp regret) and interval 4 (0.028-0.047 where 0.078 is best, 0.21-0.29 pp). No lookahead is worse at intervals 2 and 4 (0.34 / 0.29 pp) and always picks the smallest or second-smallest candidate after interval 1.

Linear decay and searched constants differ only at intervals 3 and 4, in opposite directions (linear conv-only is better at interval 3, searched constants at interval 4), both by about 0.2-0.3 pp: not resolved with one seed.

Experiment 2. Hparam importance from scratch

Question

How important is each of the four LRs when they are searched jointly from scratch, and does the answer depend on the main interval length?

Setup

Item	Value
Model / data	baseline_bs2000 except the four trained LRs (conv, head, norm_bias, whiten_bias), which are constant for the first L training steps from initialization
Main interval L	1, 2, 5, 10, 40 steps; the score is taken right after the L steps (no lookahead)
Joint search	From scratch: every LR starts at 1.0. Scan each LR over x10^k, k = -3..3 (others fixed), keep the best, snap to the 0.6^k grid, then coordinate-climb on the grid (one step up / down, move while the score improves) until no LR moves: the center
Sweep	From the center, each LR alone over 5 grid points on each side (others at the center). Offset +1 = one grid step larger (x1/0.6), -1 = one step smaller (x0.6)
Seed / score	1 seed (0); test-set TTA accuracy (%); the test cross-entropy loss is logged too

Joint-search centers

L (steps)	Center: conv, head, norm_bias, whiten_bias	Center TTA	Evaluations
1	0.13, 60, 0.00078, 60	25.43	73
2	0.22, 1, 1.7, 1300	36.89	71
5	0.078, 0.006, 2100, 1300	52.00	76
10	0.078, 0.017, 2100, 13	59.47	70
40	0.078, 2100, 280, 2100	84.60	73

Importance: TTA drop from the center (pp) at 1 step / 2 steps (mean of both sides) / worst within 5 steps

LR	L = 1	L = 2	L = 5	L = 10	L = 40
conv	0.58 / 2.01 / 16.42	1.00 / 2.90 / 19.69	3.20 / 7.96 / 32.96	2.02 / 4.11 / 29.37	1.95 / 5.26 / 49.82
head	0.03 / 0.14 / 0.29	0.02 / 0.02 / 0.08	0.10 / 0.12 / 0.18	0.32 / 0.49 / 0.67	0.54 / 1.65 / 62.56
norm_bias	0.00 / 0.00 / 0.00	0.05 / 0.02 / 0.08	0.51 / 1.46 / 28.09	1.12 / 2.77 / 28.84	0.43 / 0.84 / 6.98
whiten_bias	0.01 / 0.03 / 0.09	0.15 / 0.85 / 4.48	0.56 / 1.02 / 3.45	0.51 / 0.58 / 2.00	0.46 / 0.96 / 2.44

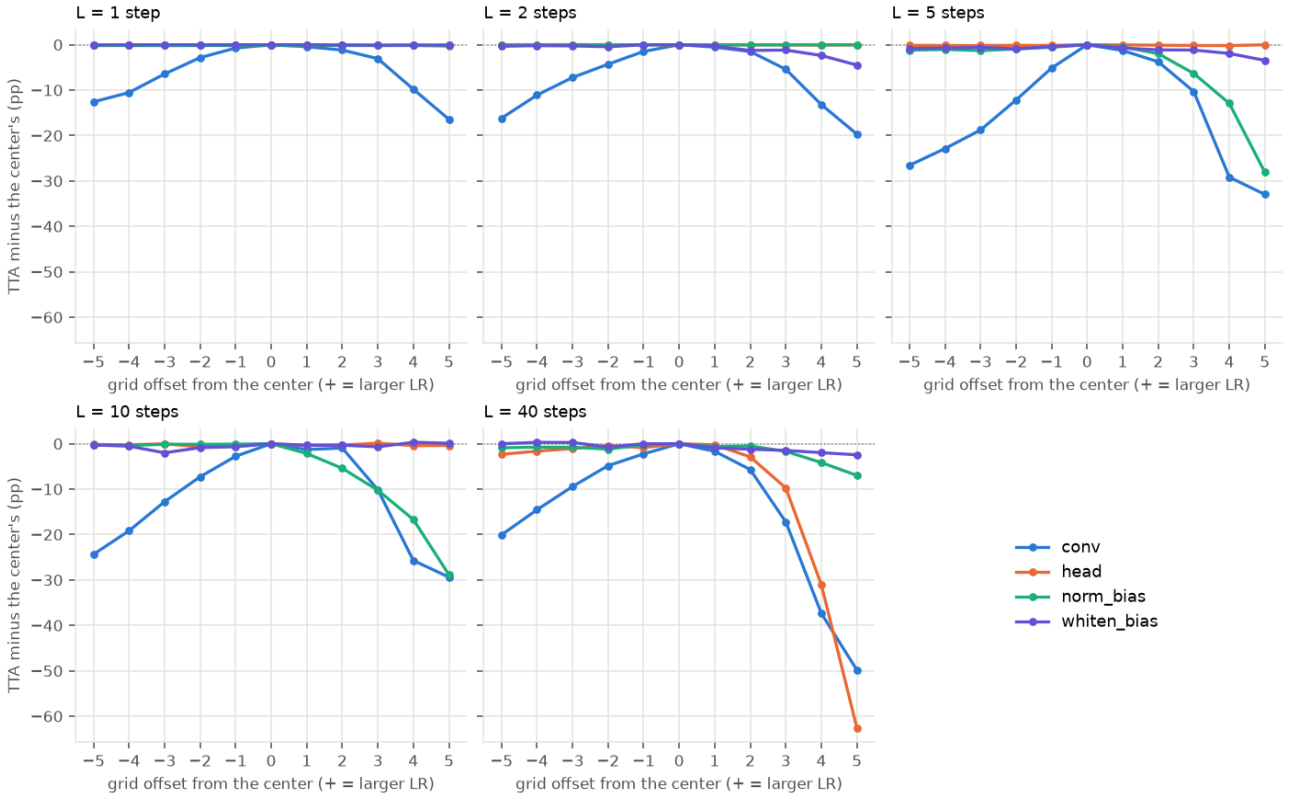


Figure 2. One-at-a-time sweeps around each L's joint-search center.

L = 1: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 25.43)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.13)	-12.58	-10.53	-6.44	-2.84	-0.73	+0.00	-0.43	-1.18	-3.06	-9.80	-16.42
head (60)	-0.15	-0.14	-0.17	-0.16	-0.03	+0.00	-0.03	-0.11	-0.16	-0.12	-0.29
norm_bias (0.00078)	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00
whiten_bias (60)	-0.03	-0.02	-0.01	-0.01	-0.01	+0.00	-0.01	-0.05	-0.09	-0.08	-0.04

L = 2: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 36.89)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.22)	-16.19	-11.09	-7.21	-4.29	-1.48	+0.00	-0.52	-1.50	-5.41	-13.19	-19.69
head (1)	+0.01	+0.01	+0.02	+0.00	-0.01	+0.00	-0.03	-0.03	-0.05	-0.08	-0.04
norm_bias (1.7)	-0.07	-0.08	-0.05	-0.01	-0.04	+0.00	-0.05	-0.03	-0.03	-0.02	-0.01
whiten_bias (1300)	-0.34	-0.20	-0.26	-0.44	-0.01	+0.00	-0.29	-1.26	-1.18	-2.37	-4.48

L = 5: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 52.00)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.078)	-26.55	-22.87	-18.78	-12.17	-5.13	+0.00	-1.28	-3.75	-10.31	-29.17	-32.96
head (0.006)	-0.13	-0.17	-0.17	-0.14	-0.16	+0.00	-0.04	-0.10	-0.17	-0.18	-0.02
norm_bias (2100)	-1.19	-1.04	-1.29	-0.94	-0.54	+0.00	-0.47	-1.97	-6.33	-12.92	-28.09
whiten_bias (1300)	-0.81	-0.75	-0.58	-0.91	-0.41	+0.00	-0.71	-1.13	-1.17	-1.92	-3.45

L = 10: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 59.47)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.078)	-24.35	-19.17	-12.77	-7.28	-2.76	+0.00	-1.28	-0.95	-10.13	-25.72	-29.37
head (0.017)	-0.17	-0.23	-0.02	-0.67	-0.31	+0.00	-0.34	-0.30	+0.13	-0.42	-0.38
norm_bias (2100)	-0.29	-0.39	-0.12	-0.13	-0.09	+0.00	-2.15	-5.42	-10.23	-16.70	-28.84
whiten_bias (13)	-0.26	-0.53	-2.00	-0.85	-0.67	+0.00	-0.36	-0.32	-0.68	+0.32	+0.11

L = 40: TTA minus the center's (pp) at each grid offset (center LR in brackets; center TTA 84.60)

LR	-5	-4	-3	-2	-1	+0	+1	+2	+3	+4	+5
conv (0.078)	-20.08	-14.54	-9.42	-4.83	-2.24	+0.00	-1.67	-5.69	-17.22	-37.36	-49.82
head (2100)	-2.33	-1.63	-1.07	-0.40	-0.85	+0.00	-0.23	-2.90	-9.74	-31.04	-62.56
norm_bias (280)	-0.90	-0.75	-0.77	-1.20	-0.24	+0.00	-0.62	-0.48	-1.69	-4.15	-6.98
whiten_bias (2100)	+0.00	+0.30	+0.27	-0.72	-0.01	+0.00	-0.91	-1.21	-1.47	-1.97	-2.44

Experiment 3. Representation quality of schedule families

Question

How close to the baseline can each family of conv LR schedules get, under no transform and under the ramp, reverse ramp and wave transforms, when searched by the same global search?

Setup

Item	Value
Model / data	baseline_bs2000; only the conv LR schedule is searched, the other groups stay on the baseline. The transform multiplies the conv LR the search specifies
Conditions	No transform; ramp x 4^(2x-1) (x0.25 -> x4); reverse ramp x 4^(1-2x) (x4 -> x0.25); wave x 2^sin(2 pi x) (x1 -> x2 -> x0.5 -> x1); x = fraction of training
Families	5 constant intervals (5 values); linear decay to 0 (initial LR); power decay lr x (1-t)^p (lr, p; p = 1 is linear); 2 knots at t = 0, 0.5; 3 knots at t = 0, 1/3, 2/3; 5 knots at t = 0, 0.2, 0.4, 0.6, 0.8 (knot values; linear between knots, 0 at the end)
Search	Generalized global_neighbour (global_family): coordinate search over the family's parameters on multiplicative grids (LR values x0.6, p x0.8), one grid step at a time, moving while the score improves; every candidate is a full training run on seed 0 from the same initial state
Start	The oracle spec (baseline conv schedule divided by the transform) projected onto the family: interval means or knot values snapped to the grid; linear / power start at 0.22 with p = 1
Evaluation	Start and best schedules retrained on held-out seeds 100-107 (mean test TTA; not part of the search)

References (held-out TTA, seeds 100-107)

	No transform	Ramp	Reverse ramp	Wave
Baseline linear decay, no transform (the target)	93.92	93.92	93.92	93.92
Baseline spec kept under the transform (do-nothing)	93.92	93.42	93.57	93.72

Held-out TTA of the best schedule (start = the projected oracle before search; search cost)

Family	No transform	Ramp	Reverse ramp	Wave
5 constant intervals	**93.62** (start 93.52; 18 runs, 182 s)	**93.65** (start 93.32; 28 runs, 285 s)	**93.86** (start 93.81; 28 runs, 279 s)	**93.82** (start 93.11; 22 runs, 222 s)
Linear decay (1 param)	**93.86** (start 93.92; 3 runs, 31 s)	**93.42** (start 93.42; 3 runs, 31 s)	**93.59** (start 93.57; 4 runs, 41 s)	**93.91** (start 93.72; 3 runs, 31 s)
Power decay (lr, p)	**93.86** (start 93.92; 8 runs, 81 s)	**93.95** (start 93.42; 14 runs, 141 s)	**93.59** (start 93.57; 6 runs, 61 s)	**93.91** (start 93.72; 5 runs, 51 s)
2 knots (t = 0, 0.5)	**93.86** (start 93.86; 5 runs, 51 s)	**93.77** (start 93.14; 9 runs, 91 s)	**93.88** (start 93.85; 8 runs, 83 s)	**93.83** (start 93.72; 12 runs, 121 s)
3 knots (t = 0, 1/3, 2/3)	**93.86** (start 93.94; 12 runs, 121 s)	**93.68** (start 93.74; 10 runs, 101 s)	**93.78** (start 93.90; 13 runs, 131 s)	**94.06** (start 93.90; 7 runs, 73 s)
5 knots (t = 0, 0.2, ..., 0.8)	**93.94** (start 93.97; 13 runs, 131 s)	**93.96** (start 93.91; 17 runs, 170 s)	**93.91** (start 93.91; 11 runs, 111 s)	**93.94** (start 93.97; 11 runs, 111 s)

Best parameter values (LR values in order; power: lr, p)

Family	No transform	Ramp	Reverse ramp	Wave
5 constant intervals	0.13, 0.22, 0.078, 0.047, 0.028	0.36, 0.13, 0.078, 0.028, 0.0036	0.028, 0.078, 0.13, 0.078, 0.047	0.078, 0.078, 0.13, 0.078, 0.017
Linear decay (1 param)	0.13	0.22	0.078	0.13
Power decay (lr, p)	0.13, 0.64	0.36, 2	0.078, 1	0.13, 1
2 knots (t = 0, 0.5)	0.22, 0.13	0.6, 0.078	0.078, 0.078	0.13, 0.078
3 knots (t = 0, 1/3, 2/3)	0.13, 0.22, 0.047	0.36, 0.13, 0.047	0.078, 0.078, 0.047	0.13, 0.078, 0.13
5 knots (t = 0, 0.2, ..., 0.8)	0.13, 0.13, 0.13, 0.078, 0.047	1, 0.36, 0.22, 0.047, 0.017	0.047, 0.078, 0.078, 0.13, 0.13	0.13, 0.078, 0.078, 0.13, 0.078

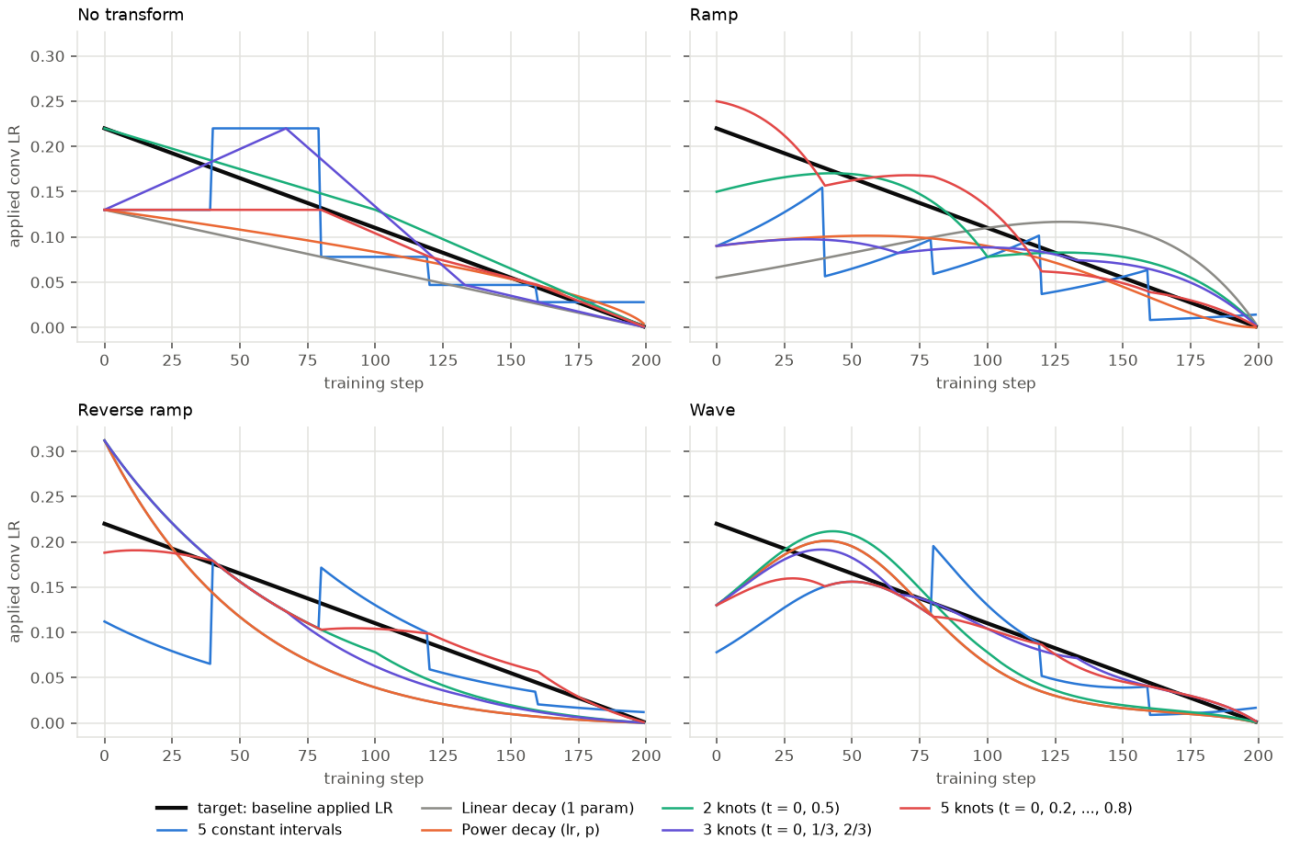


Figure 3. Applied conv LR (spec x transform) of each family's best schedule; the black line is the baseline's applied LR, which a perfect search would reproduce.

Observations

Held-out means over 8 seeds have a standard error of about 0.045 pp, so differences below about 0.1 pp are not resolved. Target: the baseline's 93.92 on the same seeds.

5 knots is the only family that reached the target in all four conditions in these searches (93.91-93.96), and its start, the oracle projected onto the family, is already there (93.91-93.97): 5 knots on the grid represent all four target schedules.

5 constant intervals of conv reach only 93.62 here without a transform (start 93.52), but that is this search's local optimum, not the family's limit. Earlier 5-interval searches at batch size 2000 (same held-out seeds) did better: conv only 93.77 (interval search; conv 0.13, 0.13, 0.078, 0.047, 0.01) and 93.72 (global_interval), and with all four LRs as 5 intervals 93.92 (interval search), matching the baseline, and 93.85 (global_interval). This search kept the last interval at 0.028 where those went to 0.01-0.017: a single-seed coordinate search can stop one grid step short at the end. Under the transforms it reaches 93.65 / 93.86 / 93.82 (ramp / reverse ramp / wave; projected starts 93.32 / 93.81 / 93.11).

Linear decay matches the target without a transform (93.86-93.92) and under wave (93.91), but no initial LR compensates a ramp (93.42) or a reverse ramp (93.59): the shape is wrong, not the scale. Power decay fixes the ramp (93.95 with p = 2, lr 0.36: decaying faster) but not the reverse ramp (93.59): (1-t)^p cannot rise early, which the reverse ramp needs.

2 and 3 knots land in between (93.68-94.06), with no consistent ordering across conditions.

Single-seed search sometimes leaves a good start for a worse held-out schedule (no transform: linear 93.92 -> 93.86, 3 knots 93.94 -> 93.86, 5 constant intervals moved its first value from 0.22 to 0.13): the search optimizes seed 0's test score, the same winner's curse as before.

Experiment 4. Hparam range from scratch (Experiment 2 redone)

Question

Experiment 2's +-5 sweeps depend on where the center landed. How far can each LR move from the center, on each side, before the score drops by 10%? A range that ends on both sides marks an LR with an optimum; a side that never drops marks an LR the score is insensitive to in that direction.

Setup

Item	Value
Model / score	As in Experiment 2: baseline_bs2000 except the four LRs, constant for the first L steps from initialization; test TTA right after the L steps; 1 seed (0)
Center	Experiment 2's joint-search center for each L (re-evaluated; it reproduces Experiment 2's TTA)
Probe	One LR at a time (others at the center), one 0.6^k grid step at a time on each side (+1 = x1/0.6, larger), until TTA <= 0.9 x the center's TTA (a 10% relative drop) or 20 steps. A diverged run counts as a drop

Stop offsets (smaller side / larger side) and the LR range within 10% of the center

LR	L = 1	L = 2	L = 5	L = 10	L = 40
conv	-2 / +3 0.078 - 0.36	-2 / +3 0.13 - 0.6	-2 / +3 0.047 - 0.22	-2 / +3 0.047 - 0.22	-3 / +3 0.028 - 0.22
head	<-20 / +7 0.0022 - 1300	<-20 / +14 3.7e-05 - 770	<-20 / >20 2.2e-07 - 170	<-20 / >20 6.1e-07 - 460	<-20 / +3 0.078 - 5900
norm_bias	<-20 / >20 2.9e-08 - 21	<-20 / +17 6.1e-05 - 5900	<-20 / +3 0.078 - 5900	<-20 / +3 0.078 - 5900	<-20 / +6 0.01 - 3500
whiten_bias	<-20 / +10 0.0022 - 5900	<-20 / +5 0.047 - 9800	<-20 / +6 0.047 - 16000	<-20 / +17 0.00047 - 46000	<-20 / +11 0.078 - 350000

<-20 / >20: no 10% drop within 20 steps on that side (the range shown then ends at the last probed point). Center TTA: L = 1: 25.43, L = 2: 36.89, L = 5: 52.00, L = 10: 59.47, L = 40: 84.60.

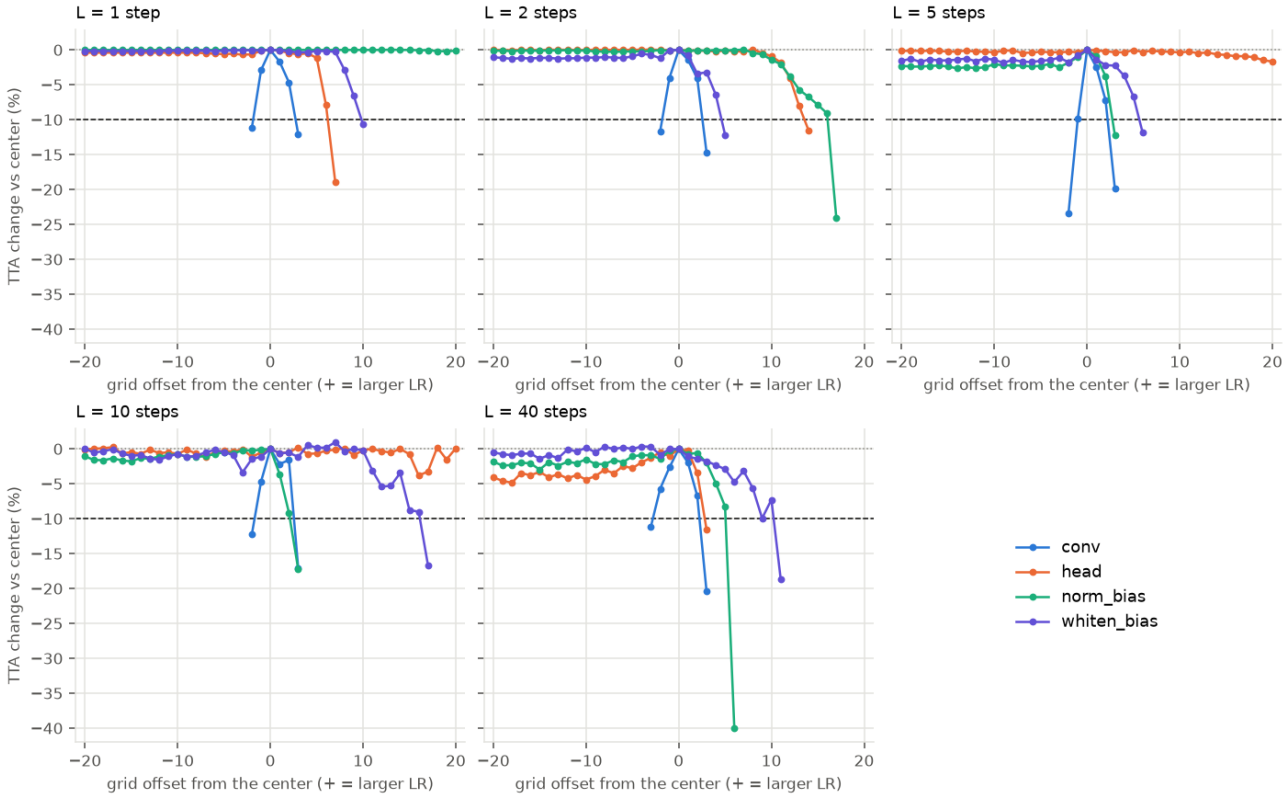


Figure 4. Relative TTA change along each probe; the dashed line is the 10% stop. Points below -40% are drawn at -40%.

Observations

One seed: single-run noise is about 0.13 pp at the full 200 steps; the stop criterion (a 10% relative drop: 2.5 pp at L = 1, 8.5 pp at L = 40) is far above it, so the stop offsets are not noise.

Conv is the only LR with an optimum: its range ends on both sides at every L, at -2 or -3 grid steps (smaller) and +3 (larger). The LR range within 10% of the center spans 5-6 grid points, a factor of 4.6-7.9 in LR (0.078-0.36 at L = 1, 0.028-0.22 at L = 40).

Head, norm_bias and whiten_bias never drop by 10% on the smaller side within 20 steps (down to 0.6^20 = 3.7e-5 x the center) at any L. The worst smaller-side loss is 4.10 pp (head, L = 40); most are below 1.5 pp, and at L = 1-2 below 0.5 pp. These three LRs only have an upper limit.

Their larger sides do end, but late and without a pattern across L: head at +7 / +14 / none / none / +3 (L = 1 / 2 / 5 / 10 / 40), norm_bias at none / +17 / +3 / +3 / +6, whiten_bias at +10 / +5 / +6 / +17 / +11. So their Experiment 2 centers mostly mark where "too large" begins, not an optimum: any value below that edge scores about the same.

At L = 10 a few probe points beat the center (whiten_bias +0.57 pp, head +0.15 pp): the joint-search center is not an exact optimum on one seed, which is the center dependence this experiment was meant to remove.

Experiment 5. Which non-conv interval schedules help

Question

The two original 5-interval searches scored 93.77 (conv only) and 93.92 (all four LRs) on held-out seeds, but their conv schedules differ in the last two intervals (0.078, 0.017 vs 0.047, 0.01). With conv fixed, do the other groups' searched intervals help, and which group?

Setup

Item	Value
Schedules	C4 = conv constant 0.13, 0.13, 0.078, 0.047, 0.01 (5 x 40 steps; the four-LR interval search's conv). Searched intervals: head 2100, 2100, 2100, 2100, 1300; norm_bias 770, 770, 770, 460, 280; whiten_bias 460, 460, 280, 280, 280. Baseline: conv 0.22 -> 0, head 1300 -> 0, norm_bias 99 -> 0 (linear over 200 steps), whiten_bias 36 -> 0 over 75 steps, then 0
Runs	Plain training runs (no search) on held-out seeds 100-107, current code; the paired difference is per seed against the first row

Held-out TTA

Conv	head	norm_bias	whiten_bias	Held-out TTA (mean +- SE)	vs first row (paired)
C4	baseline	baseline	baseline	93.95 +- 0.06	-
C4	searched	baseline	baseline	93.90 +- 0.05	-0.05 +- 0.08
C4	baseline	searched	baseline	93.77 +- 0.04	-0.18 +- 0.08
C4	baseline	baseline	searched	93.88 +- 0.05	-0.07 +- 0.07
C4	searched	searched	searched	93.92 +- 0.04	-0.03 +- 0.07
baseline (0.22 -> 0)	searched	searched	searched	93.87 +- 0.04	-0.08 +- 0.07
baseline (0.22 -> 0)	baseline	baseline	baseline	93.92 +- 0.05	-0.03 +- 0.07

For reference, the original conv-only interval search's schedule (conv 0.13, 0.13, 0.078, 0.078, 0.017, the others on the baseline) scored 93.77 on the same seeds (older code).

Observations

The other groups' searched intervals do not help. With conv fixed at C4 and the other three groups on the baseline, the held-out TTA is 93.95 +- 0.06, as high as any row. Switching all three to their searched intervals gives 93.92 (-0.03 +- 0.07 paired); switching one at a time gives head -0.05 +- 0.08, whiten_bias -0.07 +- 0.07 and norm_bias -0.18 +- 0.08 (about 2.3 SE: norm_bias's searched intervals, 770 -> 280 instead of 99 -> 0, probably hurt).

The 93.77 vs 93.92 gap between the original conv-only and four-LR searches came from conv, not from the other groups. Both conv schedules with the other groups on the baseline: 0.13, 0.13, 0.078, 0.078, 0.017 scores 93.77 and C4 (0.13, 0.13, 0.078, 0.047, 0.01) 93.95, +0.18 +- 0.05 paired (7 of 8 seeds). The four-LR rerun reproduces the original run's per-seed scores exactly, so the older 93.77 run is directly comparable.

So 5 constant intervals of conv alone do match the baseline (93.95 vs the baseline's 93.92 on these seeds) when the last two intervals are low enough (0.047, 0.01); Experiment 3's conv-only search stopped at 0.047, 0.028.

The searched non-conv intervals with the baseline's linear conv score 93.87 (-0.08 +- 0.07 vs the C4 control), close to the baseline itself: the other groups' schedules matter little with either conv schedule.

Experiment 6. Seed noise of the baseline

Question

How much does the baseline's score vary across training seeds, and how many seeds does a comparison need?

Setup

Item	Value
Runs	baseline_bs2000, plain training runs on seeds 0-127 (seed 0 is the search seed; 100-107 are the held-out seeds); the seed sets the initialization, data order and augmentation
Score	Test TTA accuracy (%) and test accuracy without TTA

Baseline across seeds

Metric	Seeds	Mean	SD (one run)	SE of the mean	Min	Max
TTA	128	93.942	0.121	0.011	93.61	94.28
No TTA	128	93.164	0.140	0.012	92.69	93.52

Averaging over seeds

Seeds per mean (n)	SD of an n-seed mean: SD/sqrt(n)	Measured (disjoint blocks of n)	Smallest difference two n-seed means resolve (95%)
1	0.121	0.121 (128 blocks)	0.34 pp
2	0.086	0.089 (64 blocks)	0.24 pp
4	0.061	0.069 (32 blocks)	0.17 pp
8	0.043	0.051 (16 blocks)	0.12 pp
16	0.030	0.041 (8 blocks)	0.08 pp
32	0.021	0.028 (4 blocks)	0.06 pp

Seed 0 (the search seed): 93.92. Held-out seeds 100-107: mean 93.92. All seeds: 93.94.

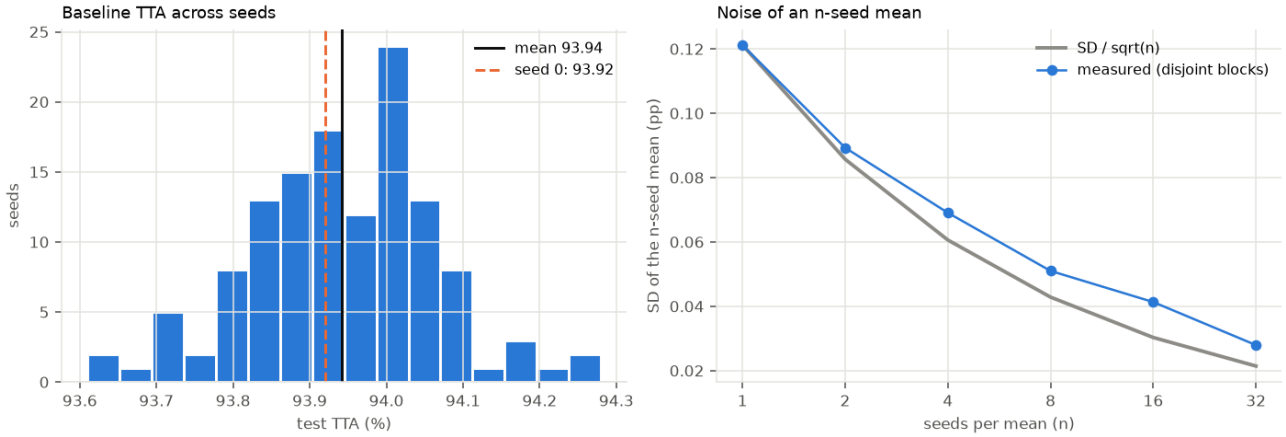


Figure 5. Left: the baseline's TTA over seeds. Right: the spread of an n-seed mean, predicted from the single-run SD and measured on disjoint blocks of seeds.

Observations

One run of the baseline has a seed SD of 0.121 pp in TTA (128 seeds: mean 93.94, range 93.61-94.28, 0.67 pp from worst to best seed); without TTA the SD is 0.140 pp (mean 93.16). The mean over all 128 seeds is known to +-0.011 pp (SE).

Averaging over seeds shrinks the noise about as SD / sqrt(n): measured on disjoint blocks of seeds, an 8-seed mean has SD 0.051 (predicted 0.043; 16 blocks, so the measured value itself is uncertain by about +-20%). Telling two schedules apart at 95% takes a difference of about 0.34 pp with 1 seed each, 0.12 pp with 8 seeds and 0.06 pp with 32 seeds (unpaired; pairing by seed helps only if the seeds' effects are shared).

The seeds used so far are typical: seed 0 (the search seed) scores 93.92 and the held-out seeds 100-107 average 93.92, both within 0.02 pp of the 128-seed mean.

Experiment 7. Hparam range around the baseline

Question

Starting from the baseline instead of a from-scratch center, how far can each LR's initial value move, on each side, before the full run's score drops by 10%?

Setup

Item	Value
Runs	baseline_bs2000, full 200-step runs, seed 0. One LR at a time (the others on the baseline)
Probe	The LR's whole baseline schedule is scaled so its initial value steps along the 0.6^k grid (+1 = x1/0.6, larger), on each side until TTA <= 0.9 x the baseline's TTA on seed 0, or 20 steps. A diverged run counts as a drop

Stop offsets (baseline TTA on seed 0: 93.92; stop at 84.53)

LR	Baseline initial LR	Stop (smaller / larger)	Initial LR range within 10%
conv	0.22	-6 / +6	0.017 - 2.8 (x165)
head	1300	<-20 / +8	0.047 - 46000 (x9.79e+05)
norm_bias	99	<-20 / +9	0.0036 - 5900 (x1.64e+06)
whiten_bias	36	<-20 / >20	0.0013 - 980000 (x7.54e+08)

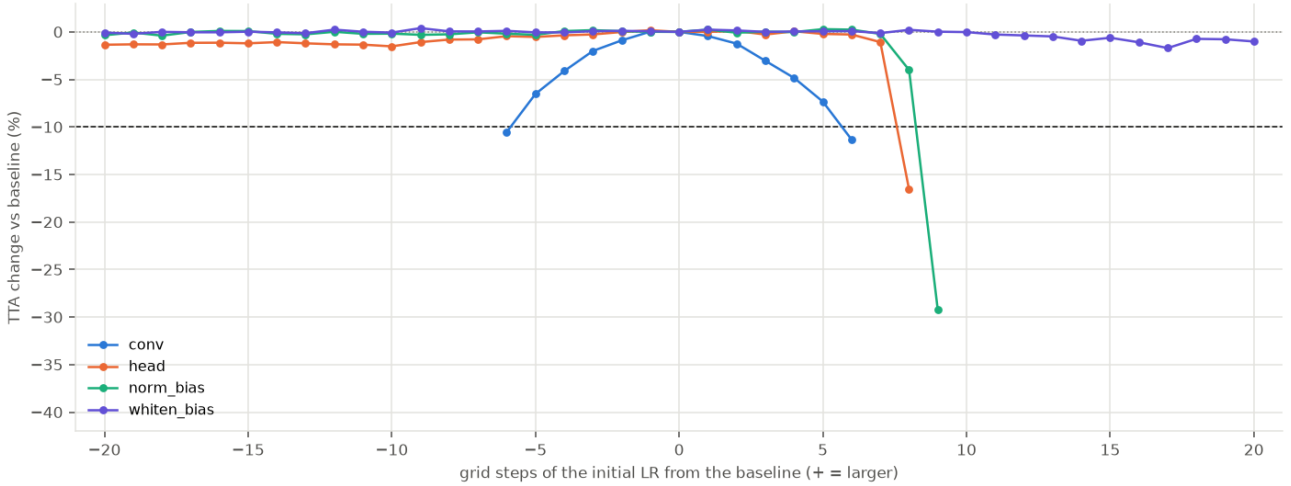


Figure 6. Relative TTA change of full runs as one LR's initial value moves from the baseline; dashed: the 10% stop. Points below -40% are drawn at -40%.

95% confidence set for the best initial LR

Each probed value is one run on seed 0, so its score carries the single-run noise of Experiment 6 (SD 0.121 pp, assumed the same at every value and independent between values). A value is kept when its score is within c of the best observed score, with c = q x SD and q the 95% quantile of max_j (Z_j - Z_1) over the k probed values (Z independent standard normals; Monte Carlo). This is Gupta's subset selection: the set contains the best probed value with probability at least 95%, even in the hardest case where all values are equally good. (open) marks an end that reaches the probe limit on a side without a 10% drop, so the set may continue beyond it.

LR	Probed points k	Cutoff c (pp)	Best observed	95% set (initial LR)	Points in set
conv	13	0.43	0.13 (93.95)	0.13 - 0.22	2 of 13
head	29	0.47	770 (94.08)	280 - 27000	10 of 29
norm_bias	30	0.47	1300 (94.19)	(open) 0.006 - 3500	22 of 30, with gaps
whiten_bias	41	0.48	0.36 (94.29)	(open) 0.0013 - 5900	28 of 41, with gaps

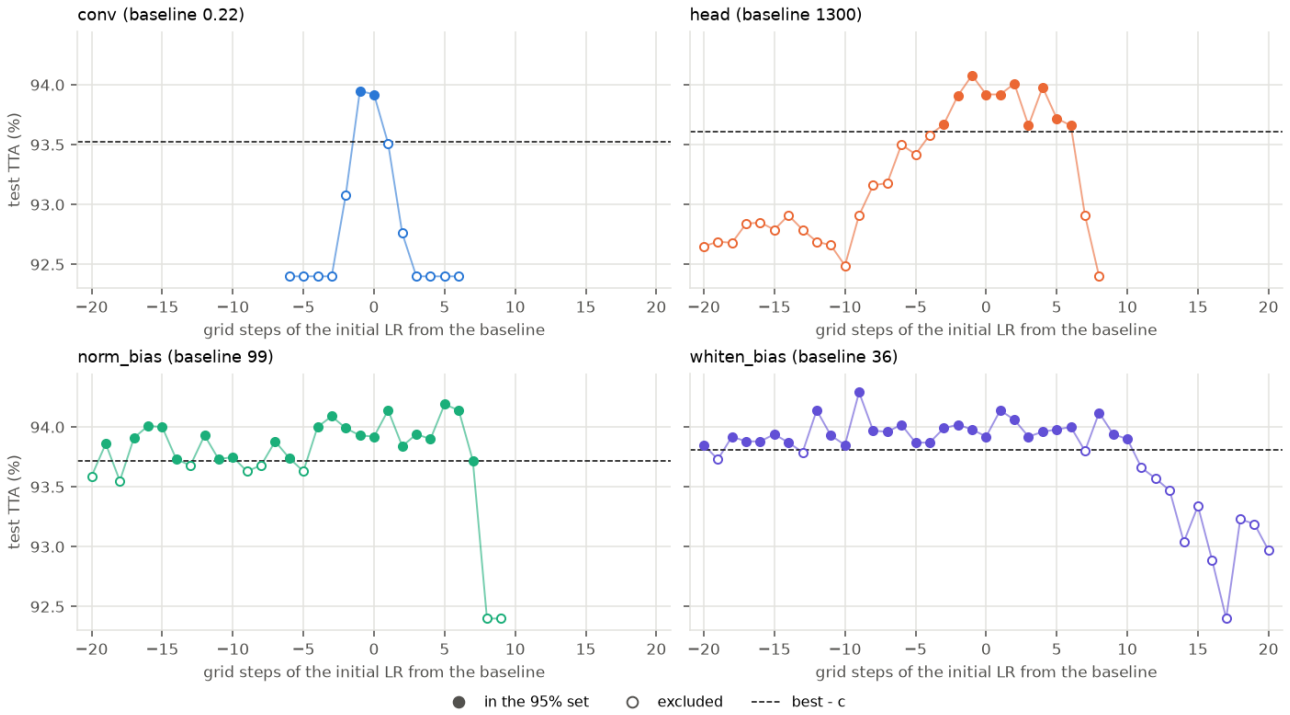


Figure 7. The probes near the top: filled points are in the 95% set (score at least best - c, dashed); points below 92.4 are drawn at 92.4.

Observations

One seed (0); the 10% stop (84.53) is about 80 single-run SDs (Experiment 6) below the baseline, so the stop offsets are not noise; differences under about 0.34 pp between single points are.

Conv has a clear optimum, and the baseline sits on it: 0.13 scores 93.95 and 0.22 (baseline) 93.92; one step further costs 0.4-0.8 pp (0.36: 93.51; 0.078: 93.08). Its 10% range ends at -6 (0.01: 84.03) and +6 (4.6: 83.28), so in full runs conv tolerates a factor of about 165 (0.017 to 2.8) before losing 10%, much wider than in the from-scratch L-step setting of Experiment 4 (5-6 grid points).

Head, norm_bias and whiten_bias never drop by 10% on the smaller side within 20 steps (down to 3.7e-5 x the baseline). Norm_bias and whiten_bias are flat there (all within 93.55-94.29, within about 0.4 pp of the baseline, roughly the single-run noise range); head loses about 1.2 pp once it is below about 8 (92.5-92.9 from 7.7 down to 0.047), still far from 10%.

On the larger side, head stops at +8 (76000: 78.35; 46000 still 92.91) and norm_bias at +9 (9800: 66.44; 5900: 90.16), both after 6-7 steps of no loss: they fail by blowing up, not gradually. whiten_bias never stops: up to 980000 (x27000) it stays above 92.3.

So around the baseline only conv needs tuning; the three other LRs only need to stay below a blow-up edge about 35x above their baseline values (head, norm_bias; the last points without loss are 46000 and 3500), and whiten_bias barely matters at all.

95% confidence sets for the best initial LR (single runs with SD 0.121 pp; cutoff c = 0.43-0.48 pp below the best observed score, wider for more probed points). Conv: 0.13-0.22, the two values at the top (93.95, 93.92); the next ones are excluded (0.36: 93.51, only 0.01 pp under the cutoff 93.52; 0.078: 93.08), so the best conv initial LR is pinned to two grid points, and the baseline's 0.22 is one of them.

Head: 280-27000 (10 of 29 points, contiguous), a factor of about 100 around the best observed 770; below 280 the scores fall off (170: 93.58), and 46000 is 1.0 pp below the baseline (92.91). The baseline's 1300 is inside.

Norm_bias: anything from the smallest probed values (0.006; open, as the probe never dropped 10% on that side) up to 3500, with gaps: the scores below 3500 are within noise of each other, so the data do not locate a best value, only an upper edge. Whiten_bias: likewise from 0.0013 (open) up to 5900; from 9800 upward the scores fall (93.66 and lower) and are excluded.

Experiment 8. Largest LR within the margin, per L (from Experiment 4)

Question

The interval search picks each LR in its first interval as the largest LR scoring within a margin of the best (directional_search with initial_lr_search, initial_dropoff_margin = 0.02). Applied at main interval lengths L = 1, 2, 5, 10, 40 from scratch, does the pick at L = 1 predict the pick at L = 40?

Setup

Item	Value
Data	Experiment 4's probes (no new runs): for each L, one LR at a time on the 0.6^k grid around Experiment 2's center, the other three LRs at the center, TTA right after L steps, seed 0. The probes extend past every point where the rule stops
Rule	As in train_cifar.py: from the center, step up while the score stays within the margin of the best so far, then down the same way (at most 20 steps per side); pick the largest LR within the margin of the best. Margin 0.02 TTA = 2 pp (the code's value); 1 and 0.5 pp shown too

Picks with the code's margin (2 pp): LR (grid steps from the center), and where it lies relative to Experiment 7's full-run 95% range for the best initial LR (last row)

L (steps)	Center TTA	conv	head	norm_bias	whiten_bias
1	25.43	0.36 (+2), above	770 (+5), in	21 (+20)*, in	5900 (+9), in
2	36.89	0.6 (+2), above	460 (+12), in	770 (+12), in	5900 (+3), in
5	52.00	0.13 (+1), in	170 (+20)*, below	5900 (+2), above	9800 (+4), above
10	59.47	0.22 (+2), in	36 (+15), below	2100 (+0), in	2100 (+10), in
40	84.60	0.13 (+1), in	3500 (+1), in	1300 (+3), in	9800 (+3), above
Exp. 7	full run	0.13 - 0.22	280 - 27000	(open) 0.006 - 3500	(open) 0.0013 - 5900

* the upward probe reached the 20-step limit without falling out of the margin, so the pick is the probe limit. in / above / below: inside, above or below the Experiment 7 range ((open): that end of the range is not bounded by the data). Experiment 7 is a different context (full 200-step runs, the other LRs on the baseline schedules), the target a short-horizon pick should predict.

Conv pick by margin

L (steps)	conv, margin 2 pp	conv, margin 1 pp	conv, margin 0.5 pp
1	0.36, above	0.22, in	0.22, in
2	0.6, above	0.36, above	0.22, in
5	0.13, in	0.078, below	0.078, below
10	0.22, in	0.078, below	0.078, below
40	0.13, in	0.078, below	0.078, below
Exp. 7	0.13 - 0.22	0.13 - 0.22	0.13 - 0.22

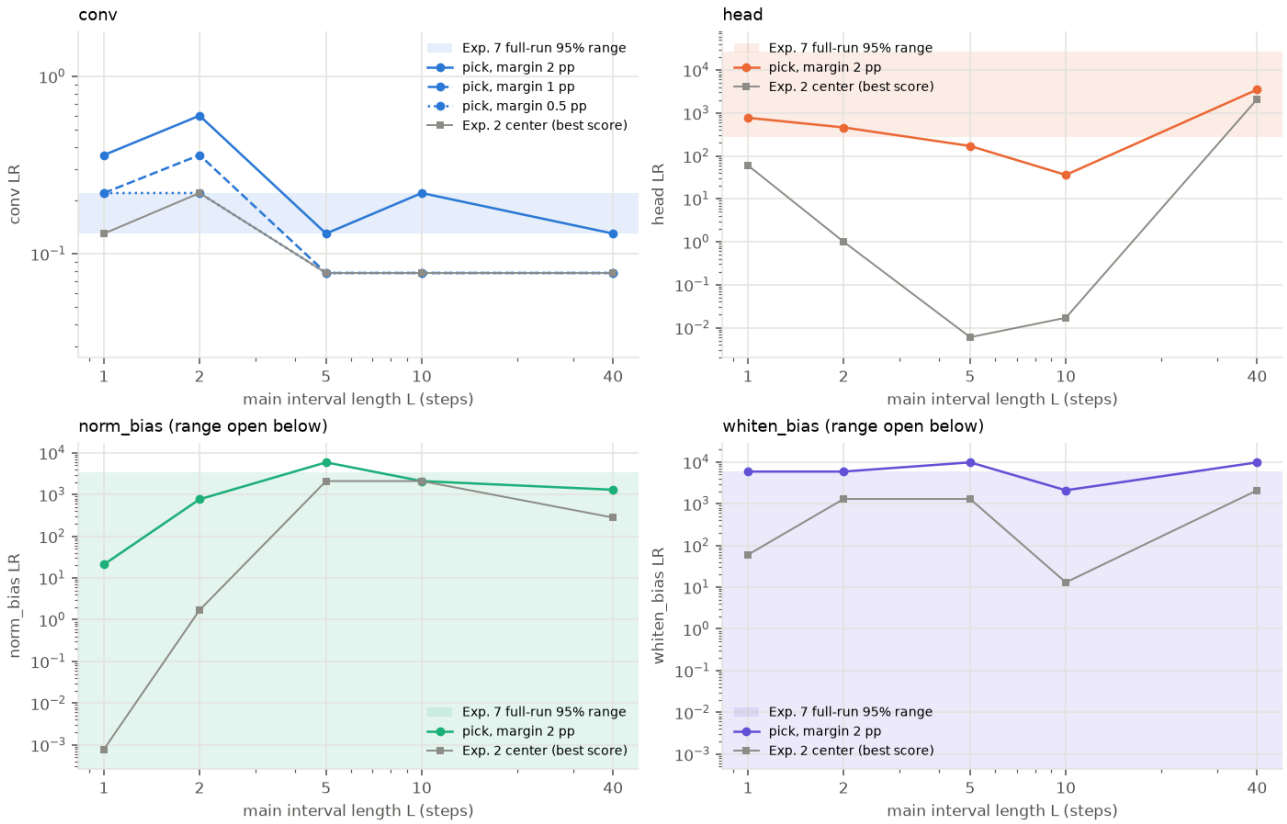


Figure 8. The rule's pick per L (conv at three margins, the others at the code's 2 pp), Experiment 2's best-score center, and the shaded reference: Experiment 7's full-run 95% range for the best initial LR.

Observations

Reference: Experiment 7's full-run 95% ranges for the best initial LR (conv 0.13-0.22, head 280-27000, norm_bias up to 3500, whiten_bias up to 5900; the last two open below, so for them only "above" is a real miss). A good short-horizon pick lands inside.

Conv: with the code's 2 pp margin, L = 1 and L = 2 pick above the range (0.36, 0.6) and L = 5, 10 and 40 inside it (0.13, 0.22, 0.13). So L = 1 does not predict L = 40's pick (2 grid steps apart, 0.36 vs 0.13), and the long-horizon picks are the ones that match the full run. With tighter margins it reverses: at 1 pp and 0.5 pp, L = 1 picks 0.22 (inside) and L = 5-40 pick 0.078 (one step below the range). No single margin puts every L inside.

Why the margin interacts with L: it is absolute, but the score scale is not (the centers score 25.43 at L = 1 and 84.60 at L = 40), and at every L the 2 pp pick sits 1-2 grid steps above that L's best-score center (L = 1: 0.13 -> 0.36; L = 40: 0.078 -> 0.13). The best-score centers alone are inside the range at L = 1 and 2 (0.13, 0.22) and one step below it at L >= 5 (0.078): short horizons favour slightly larger conv LRs than long ones, and the 2 pp rule's upward step compensates at L >= 5 but overshoots at L <= 2.

Head: L = 1 (770), 2 (460) and 40 (3500) pick inside the range; L = 5 (170) and 10 (36) below it, by 1-4 grid steps although the upward probe at L = 5 hit the 20-step limit. norm_bias and whiten_bias: their ranges are open below, so the only misses are picks above the upper edge: norm_bias at L = 5 (5900 > 3500), whiten_bias at L = 5 and 40 (9800 > 5900), each one grid step over. L = 1's picks (21 and 5900) are inside, but their ranges are wide enough that this says little.

So in this data L = 1 lands inside the full-run range for all three of head, norm_bias and whiten_bias (L = 40: two of three), but for conv, the one LR with a narrow range, L = 1 overshoots with the code's margin, while L >= 5 lands inside.

Caveat: in each L the other three LRs sit at that L's own Experiment 2 center, which differs between L (e.g. head 60 at L = 1, 0.006 at L = 5, 2100 at L = 40), so the comparison across L is not in a fixed context, and Experiment 7's context (full runs, baseline schedules) differs from all of them. Running the rule from a common start at every L would remove the first confound; it needs new runs.

Experiment 9. Representation quality of schedule families, noise-aware search

Question

Experiment 3's search scored every candidate on one run (seed 0), so it climbed on seed noise and stopped at schedules that were lucky on seed 0 (e.g. 5 constant intervals: 93.99 on seed 0, 93.62 held-out). With a search that only moves to a neighbour that is statistically better given the seed noise, and adds seeds until it can tell, how close does each family get?

Setup

Item	Value
As Experiment 3	baseline_bs2000, conv only (the others on the baseline); the same families, conditions (no transform, ramp, reverse ramp, wave), grids (LR x0.6, p x0.8) and starts (the oracle projected onto the family); start and final retrained on held-out seeds 100-107
Search	Global neighbour (noisy_family): each round compares the current schedule with all its neighbours (each parameter one grid step up or down; k = 2 for linear, 4 for power and 2 knots, 6 for 3 knots, 10 for 5 intervals and 5 knots). Every point's score is its mean TTA over full runs on search seeds 0, 1, 2, ...
Decision	d = neighbour mean - current mean, SE = 0.12 pp x sqrt(1/n_neighbour + 1/n_current) (the single-run SD of Experiment 6, runs treated as independent). One-sided 80% bounds, Bonferroni over the k neighbours: z = 1.28 / 1.64 / 1.83 / 2.05 for k = 2 / 4 / 6 / 10. Better: d - z SE > 0. Ruled out: d + z SE < 0. Otherwise undecided
Moves and seeds	If any neighbour is better, move to the better one with the highest mean. Else run one more seed on the current schedule and on every undecided neighbour, and repeat. Stop when every neighbour is ruled out, or at 8 seeds per point (undecided neighbours then count as not better)

Held-out TTA of the final schedule (Experiment 3's for comparison; the start; search cost)

Family	No transform	Ramp	Reverse ramp	Wave
5 constant intervals	running	running	running	running
Linear decay (1 param)	**93.92** (Exp. 3: 93.86; start 93.92; 0 moves, 15 runs)	**93.42** (Exp. 3: 93.42; start 93.42; 0 moves, 5 runs)	running	running
Power decay (lr, p)	**93.97** (Exp. 3: 93.86; start 93.92; 1 moves, 36 runs)	**93.70** (Exp. 3: 93.95; start 93.42; 1 moves, 21 runs)	running	running
2 knots (t = 0, 0.5)	**94.03** (Exp. 3: 93.86; start 93.86; 1 moves, 38 runs)	running	running	running
3 knots (t = 0, 1/3, 2/3)	**93.94** (Exp. 3: 93.86; start 93.94; 0 moves, 48 runs)	running	running	running
5 knots (t = 0, 0.2, ..., 0.8)	running	running	running	running

Target: the baseline's 93.92 on the held-out seeds; do-nothing references as in Experiment 3 (ramp 93.42, reverse ramp 93.57, wave 93.72).

Final parameter values (seeds behind the final mean; why the search stopped)

Family	No transform	Ramp	Reverse ramp	Wave
5 constant intervals	running	running	running	running
Linear decay (1 param)	0.22 (7 seeds; none can be better)	0.22 (2 seeds; none can be better)	running	running
Power decay (lr, p)	0.22, 1.2 (8 seeds; max seeds)	0.22, 1.2 (7 seeds; none can be better)	running	running
2 knots (t = 0, 0.5)	0.22, 0.078 (8 seeds; none can be better)	running	running	running
3 knots (t = 0, 1/3, 2/3)	0.22, 0.13, 0.078 (8 seeds; max seeds)	running	running	running
5 knots (t = 0, 0.2, ..., 0.8)	running	running	running	running

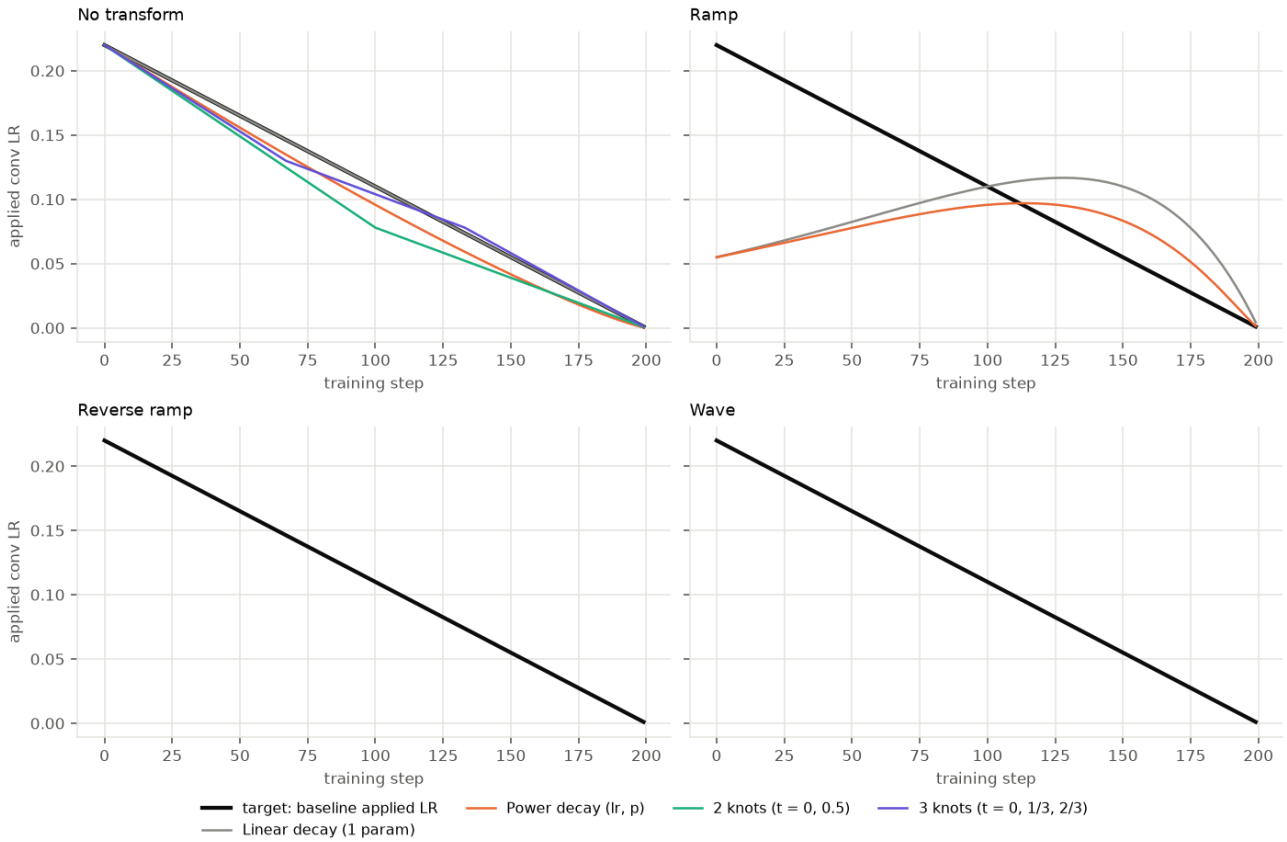


Figure 9. Applied conv LR of each family's final schedule (noise-aware search); the black line is the baseline's applied LR.